LESSON 3.6

Ratio Reasoning: Convert Measurements within the Same System

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use ratio reasoning to convert between units within the same measurement system.

WHY THIS MATTERS Chef Reyes writes every recipe in the units the supplier uses, but the kitchen's measuring tools are marked in different units of the same system. A stock order lists quarts; the pot is marked in cups. A height rule at the county fair ride is posted in inches; the sign-up sheet lists feet. Your job is to convert between units without changing the amount. The stock order lists 3 quarts, but the pot is marked in cups. The height rule at the ride says 48 inches, but the sign-up sheet lists feet.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Conversion factor *Spanish: Factor de conversión*	The ratio that links two units in the same measurement system, such as 12 inches to 1 foot. Within one system it is exact.	1 ft = 12 in
Unit ratio *Spanish: Razón unitaria*	A ratio that compares an amount to exactly 1 unit of another quantity, like 12 inches to 1 foot.	12 : 1 — twelve inches for every one foot
Convert *Spanish: Convertir*	To change a measurement from one unit to another without changing how much it is.	6 ft → 72 in
Measurement system *Spanish: Sistema de medidas*	A family of units that go together — customary units like inches and feet, or metric units like milliliters and liters.	Customary: 12 in = 1 ft · Metric: 1,000 mL = 1 L
Double number line *Spanish: Recta numérica doble*	Two number lines lined up so matching tick marks show two units for the same amount.	Liters 0, 1, 2, 3, 4 above milliliters 0, 1,000, 2,000, 3,000, 4,000
Ratio table *Spanish: Tabla de razones*	A table of equal ratios that lets you scale a conversion up or down one row at a time.	1 ft : 12 in, 2 ft : 24 in, 4 ft : 48 in

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A pitcher holds 2 quarts. How many fluid ounces? (1 qt = 4 cups, 1 cup = 8 fl oz)

1. Quarts to cups: 2 × 4 = 8 cups.
2. Cups to fl oz: 8 × 8 = 64 fl oz.
3. Check: 64 ÷ 8 ÷ 4 = 2 quarts.

Answer: 2 quarts = 64 fluid ounces

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Diamond wants to drink 4 liters of water. Her bottle is marked in milliliters.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I know 1 _____ = _____ _____ .
- I _____ because I am going from a _____ unit to a _____ unit.
- My answer is _____ _____ .
- This is reasonable because _____ .

Word bank convert | conversion factor | customary | multiply | divide | larger | smaller | inch | foot | yard | pound | gallon

WATCH OUT!

A common mistake

A common mistake when converting within a measurement system is choosing the operation by habit instead of by unit size

LESSON 3.6

Ratio Reasoning: Convert Measurements within the Same System

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

There are 4 cups in 1 quart. Complete the conversion table.

Quarts	Cups
1	4
3	
6	

HINT Row 1 is the unit ratio: 1 quart is 4 cups.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

There are 12 inches in 1 foot. How many inches are in 5 feet?

(A) 7 inches
(B) 17 inches
(C) 50 inches
(D) 60 inches

HINT Write the unit ratio first: 12 inches : 1 foot.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

There are 1,000 milliliters in 1 liter. How many milliliters are in 3 liters?

(A) 30 mL
(B) 300 mL
(C) 1,003 mL
(D) 3,000 mL

HINT The unit ratio is 1,000 mL : 1 L.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

There are 16 ounces in 1 pound. How many pounds is 48 ounces?

(A) 3 pounds
(B) 32 pounds
(C) 64 pounds
(D) 768 pounds

HINT Which unit is larger, an ounce or a pound?

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A child is 4 feet tall. A ride requires 44 inches. Can the child ride?

(A) No, because 4 is less than 44

(B) Yes, because 4 feet = 44 inches exactly

(C) Yes, because 4 feet = 48 inches, which is more than 44

(D) You cannot compare feet and inches

HINT You cannot compare two different units directly.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Find the error

1. **Setup** Convert 5 feet to inches.
2. **Step 1** There are 12 inches in 1 foot.
3. **Step 2** 5 ÷ 12 = 0.42
4. **Step 3** So 5 feet is about 0.42 inches.

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Is an inch bigger or smaller than a foot?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Why does converting to a smaller unit always make the NUMBER bigger, even though the amount stays exactly the same?

SENTENCE STARTER When the unit gets ___, it takes ___ of them to make the same amount, so I ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 3.6

Ratio Reasoning: Convert Measurements within the Same System

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
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- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

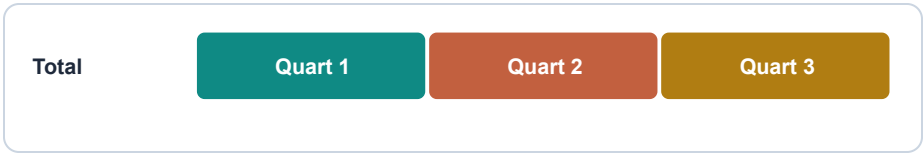
1 **Complete the table. Show how you found each value.**

There are 1,000 milliliters in 1 liter. Complete the double number line as a table.

Liters	Milliliters
1	1,000
2	
4	

SHOW YOUR WORK

2 **Use the model to solve. Show your work.**



A 3-quart stockpot is filled using a 1-cup scoop. There are 4 cups in 1 quart. Use the bar model to show how many scoops fill the pot.

SHOW YOUR WORK

Answer: _____

3 **Write the name of the correct group on each line.**

Sort each conversion by what happens to the NUMBER.

Groups: **The number gets bigger** **The number gets smaller**

_____ 7 feet → inches

_____ 84 inches → feet

_____ 2 liters → milliliters

_____ 5,000 milliliters → liters

_____ 3 pounds → ounces

_____ 64 ounces → pounds

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

Diamond needs 4 liters of water. Her bottle holds 500 milliliters. How many times must she fill it?

(A) 2 times
(B) 4 times
(C) 8 times
(D) 40 times

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Chef Reyes's stock order lists 3 quarts of broth, but her measuring cup is marked in cups. There are 4 cups in 1 quart. How many cups of broth is that?

(A) 7 cups
(B) 0.75 cup
(C) 12 cups
(D) 3 cups

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Find the error

1. **Setup** A recipe needs 2 pounds of flour. The scale reads in ounces (16 oz = 1 lb).
2. **Step 1** The unit ratio is 16 ounces : 1 pound.
3. **Step 2** 2 × 16 = 32, so 2 pounds = 32 ounces.
4. **Step 3** The scale reads 40 ounces, so there is not enough flour.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Why does converting to a smaller unit always make the NUMBER bigger, even though the amount stays exactly the same?

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Challenge · Extension

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PART 1 Understand the idea

models and meaning

1 **Decide whether each pair balances. Check Yes or No, then explain.**

1 foot / 12 inches = 7 feet / x inches → x = 84 inches

Left side	Right side	Balanced?

EXPLAIN YOUR CHOICE

_____

_____

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the error

1. **Setup** Convert 4 liters to milliliters, then find how many 250-mL cups it fills.
2. **Step 1** 4 × 1,000 = 4,000 mL
3. **Step 2** 4,000 × 250 = 1,000,000
4. **Step 3** It fills 1,000,000 cups.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

Michael's mother is setting four fence posts. Each post needs 250 ounces of concrete mix, one bag holds 80 ounces, and each bag needs 1 quart of water. How many bags and how many cups of water does she need? Show your conversions.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Why does converting to a smaller unit always make the NUMBER bigger, even though the amount stays exactly the same?

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